



Final – MATH 346/377 – Winter 2025

DATE: Tuesday, April 15 TIME: 9:00AM – 12:00PM

EXAMINER: Romain Branchereau

ASSOCIATE EXAMINER: Henri DARMON

First name (please write as legibly as possible within the boxes)

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Last name

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Student ID number

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- There are 11 problems, each worth 10 points. For Math 346, only your 8 best problems will count, for a total of 80 points. For Math 377, your 9 best problems will count, for a total of 90 points.
- Notes, calculators and other electronic devices are not permitted.



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This page will not be graded and can be used as scrap paper.



1. (10 points) True or false? Justify your answer.
 - (a) Let p be an odd prime. If g is a primitive root modulo p , then it is not a quadratic residue modulo p .
 - (b) $x = -\frac{1}{3}$ is a 3-adic *integer* (in $\mathbb{Z}_3$).
 - (c) Let p be a prime and a an integer. If $\gcd(a, p^2) = p$ then $\gcd(a + p, p^2) = p$.
 - (d) There exists an integer n for which the system $7x + (4n)y = 13$ has no solutions $x, y \in \mathbb{Z}$.



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2. (10 points) Show that the equation

$$g(x) = (x^2 - 5)(x^2 - 41)(x^2 - 205)$$

has no integer solutions, but the congruence $g(x) \equiv 0 \pmod{p}$ has a solution for every prime p .



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3. (10 points) Find all integers $0 \leq x \leq 24$ that satisfy

$$x^3 \equiv 13 \pmod{25}.$$



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4. (a) (6 points) Find all residues of order 5 modulo 31.
(b) (4 points) Deduce all integers $1 \leq x \leq 30$ solving

$$1 + x + x^2 + x^3 + x^4 \equiv 0 \pmod{31}.$$



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5. (10 points) What are the last two digits of $(2357)^{320}$? Justify your answer.



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6. (10 points) Find all integers $0 \leq x \leq 92$ that solve the equation

$$31x^{30} + 5x - 3 \equiv 0 \pmod{93 = 3 \cdot 31}.$$



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7. (a) (5 points) For which primes p is $p^2 + 1$ prime?
(b) (5 points) For which primes p is $p^2 + 2$ prime?



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8. (10 points) Use Euclid's idea to prove (by contradiction) that there are infinitely many primes congruent to 3 modulo 4.



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9. (a) (5 points) Let g be a primitive root modulo a prime $p \equiv 1 \pmod{4}$. Show that $g^{\frac{p-1}{4}}$ is a root of $x^2 + 1 \equiv 0 \pmod{p}$.
- (b) (5 points) Use part (a) to find the two roots of $x^2 + 1 \equiv 0 \pmod{13}$.



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10. (a) (8 points) Show that the Euler ϕ -function satisfies $\phi(n^2) = n\phi(n)$ for any integer $n \geq 1$. Deduce that for $k \geq 2$ and p a prime

$$\phi(\phi(p^k)) = p^{k-2}\phi((p-1)^2).$$

- (b) (2 points) Compute $\phi(\phi(5^3))$.



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11. For an odd prime p we consider the sum

$$S(p) := \sum_{k=1}^{p-1} \left(\frac{k}{p} \right).$$

- (a) (4 points) For $p = 5$, verify that $S(5) = 0$.
- (b) (6 points) Show that $S(p) = 0$ for all odd primes.



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